Lab 10
Course: Networks System Design
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Due Date: Due: Tuesday, 13 January 2026, 12:00 AM

Part 1: The Link State Walkthrough (Dijkstra)

1. The Topology

```
          (5)                          (5)
    B ---------- D -------------- F
    | \          | \              |
    |  \ (2)     |  \ (1)         |
   (2)  \        |   \            | (1)
    |    \      (2)   \           |
    |     \      |     \          |
    A ---------- C -------------- E
           (1)              (4)
```

2. The Calculation Table

Step	N' (Nodes added)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(F), p(F)
0	A	2, A	1, A	∞	∞	∞
1	A, C	2, A	1, A	3, C	5, C	∞
2	A, C, B	2, A	1, A	3, C	5, C	∞
3	A, C, B, D	2, A	1, A	3, C	4, D	8, D
4	A, C, B, D, E	2, A	1, A	3, C	4, D	5, E
5	A, C, B, D, E, F	2, A	1, A	3, C	4, D	5, E

3. The Result

- Shortest path A → F:
- A → C → D → E → F
- Total cost: 5

- Next hop from A to reach F: C

Part 2: The "Count-to-Infinity" Simulation (Distance Vector)

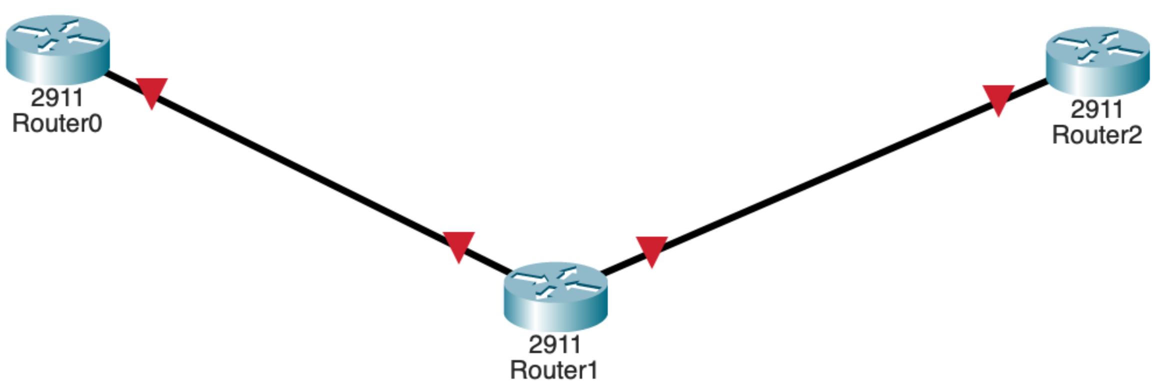
State	Y's Table (Cost to Z)	X's Table (Cost to Z)
Initial	Cost: 1 (Direct)	Cost: 5 (Next Hop: Y)
Link Change	Detects new cost 60. Checks X's advertisement: “X can reach Z with cost 5”	(Has not received update yet) Cost: 5
Iteration 1	Updates route via X: 4 (to X) + 5 (X to Z) = 9. Y tells X: “I can reach Z with cost 9”	Receives Y's update (9). Updates own cost: 4 (to Y) + 9 (Y to Z) = **13**
Iteration 2	Receives X's update (13). New Cost: 4 + 13 = **17**	Receives Y's update (17). New Cost: 4 + 17 = **21**

Question: Why is this behavior called "Bad news travels slow"?

- Answer: Because when the link Y–Z becomes very expensive (cost 60), that bad news does not spread immediately in a distance-vector protocol. Each router only knows what its neighbor advertises, so X and Y keep telling each other apparently “good” routes and gradually increase the cost step-by-step (5 → 9 → 13 → 17 → 21 → …) instead of jumping directly to 60 (or infinity).

Part 3: Packet Tracer Activity (RIP)

1. Setup Topology



2. IP Configuration

- Router0 (G0/0): 192.168.1.1 255.255.255.0

IP Configuration	
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0

- Router1 (G0/0): 192.168.1.2 255.255.255.0

IP Configuration	
IPv4 Address	192.168.1.2
Subnet Mask	255.255.255.0

- Router1 (G0/1): 192.168.2.1 255.255.255.0

IP Configuration	
IPv4 Address	192.168.2.1
Subnet Mask	255.255.255.0

- Router2 (G0/0): 192.168.2.2 255.255.255.0

IP Configuration	
IPv4 Address	192.168.2.2
Subnet Mask	255.255.255.0

- Router2 (Loopback0): 10.10.10.1 255.255.255.0

```
Router(config)#interface loopback 0

Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

Router(config-if)#ip address 10.10.10.1 255.255.255.0
```

3. Enable RIP Routing

- Router0

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#
```

- Router1

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
```

- Router2

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
```

4. Verification

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
R       10.10.10.0/24 [120/2] via 192.168.1.2, 00:00:02, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 192.168.1.2, 00:00:02, GigabitEthernet0/0

Router#
```

In the RIP routing entry `[120/2]`, what does the **2** represent?

Answer

The **2** represents the **hop count** to reach the destination network.

A hop count is the number of routers (or "hops") that a packet must travel through to get from the source router to the destination network.
In this case, a value of **2** means:

- There are **two intermediate routers** between Router0 and the destination network.
- The packet will pass through two routing devices before arriving at its final network.
- This value increases each time a route is forwarded through another RIP-enabled router.

Since RIP is a **distance-vector protocol**, hop count is the metric it uses to determine the best path.
The lower the hop count, the better the route.

So simply: **Hop Count (2) = Router0 → Router1 → Router2 → Destination**

```
Router#traceroute 10.10.10.1
Type escape sequence to abort.
Tracing the route to 10.10.10.1

  1   192.168.1.2     0 msec    0 msec    0 msec
  2   192.168.2.2     0 msec    0 msec    0 msec
Router#
```